MAIN PROJECT REPORT

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Title: Comprehensive Analysis of Fragrances in the E-Commerce Market

Project Domain: Sales & ECommerce

Project Tools Used: Python and Power BI

Submission Date: 29 Apr 2026

Mentor: Kumaran

Raw Data link:

https://raw.githubusercontent.com/Meganathan1294/Main-project-/refs/heads/main/ebay_mens_perfume.csv

https://raw.githubusercontent.com/Meganathan1294/Main-project-/refs/heads/main/ebay_womens_perfume.csv

Dataset Links:

Cleaned dataset link:

https://raw.githubusercontent.com/Meganathan1294/Main-project-/refs/heads/main/cleaned_fragrance_data%20(2).csv

Collab link:

https://colab.research.google.com/drive/1b2Rkqa5Z0UZNFpBjLrXj51xwyZAFbb_f?usp=sharing

Comprehensive Analysis of Fragrances in the E-Commerce Market

Objective

1. To identify meaningful patterns, trends, and insights from the dataset using exploratory data analysis (EDA)
2. To clean, transform, and preprocess the data for analysis
3. To create meaningful visualizations that clearly explain the story behind the data
4. To provide actionable insights and recommendations from the analysis

Outcome

The project helps uncover key patterns, trends, and hidden insights from the dataset. The analysis provides a clear understanding of the data and supports decision-making with evidence-based findings.

Dataset Information

Source: https://www.kaggle.com/datasets/kanchana1990/perfume-e-commerce-dataset-2024?resource=download

Year / Timeline: Data collected during 2024

Dataset Description: The Perfume E-Commerce Dataset 2024 comprises detailed information on 2000 perfume listings sourced from eBay, split into two separate CSV files for men's and women's perfumes, each containing 1000 entries. This dataset provides a

comprehensive view of the current market trends, pricing, availability, and geographical distribution of perfumes in the e-commerce space.

Column description

- **Brand:** The brand of the perfume.
- **Title:** The title of the listing.
- **Type:** The type of perfume (e.g., Eau de Parfum, Eau de Toilette).
- **Price:** The price of the perfume.
- **Price With Currency:** The price with currency notation.
- **Available:** The number of items available
- **Available Text:** Text description of availability.
- **Sold:** The number of items sold.
- **Last Updated:** The last updated timestamp of the listing.
- **Item Location:** The location of the item.

Type of Analysis

1. Descriptive Analysis (Summarizing the Dataset)

This level summarizes the "what" of the data based on total listings

(Men's, Women's).

- Pricing: Men's perfumes are generally more expensive, with an average price of compared to for Women's.
- Sales Volume: The Men's category shows significantly higher total sales activity on eBay (units sold) compared to the Women's category (units sold).
- Market Share: Calvin Klein and Versace are the high-volume leaders across both datasets, accounting for over units sold combined.
- Product Types: The inventory is primarily split between Eau de Parfum (EDP) and Eau de Toilette (EDT), with EDPs commanding a higher price premium.

2. Diagnostic Analysis (Finding Reasons Behind Patterns)

This level investigates "why" certain trends exist in the marketplace.

- Concentration vs. Price: Eau de Parfum listings have an average price of higher than Eau de Toilette (
- The "Mass-Appeal" Brand Effect: Brands like Calvin Klein and Davidoff have lower-than-average prices (approx.
- Gender Sales Gap: The higher sales volume in Men's perfumes despite higher average prices suggests a more "utilitarian" or "repeat-purchase" behavior in the men's segment on eBay, or a higher concentration of popular "blue" fragrances that sell in bulk.

3. Predictive Analysis (Forecasting Trends)

- Sales vs. Price Forecast: Based on current trends, if a perfume's price is increased from $\$30$ to $\$50$, the predicted sales volume per listing drops from approximately 83 units to 65 units
- Inventory Outcome: Listings for "Unknown" or "Other" fragrance types are predicted to have lower turnover rates, as they carry the highest average price ($\$58.52$) but lack the categorical clarity buyers search for
- Brand Growth: Given its high listing count (53) and high sales ($128,000+$), Versace is predicted to remain the most stable "premium-midrange" brand for consistent revenue

4. Prescriptive Analysis (Recommendations for Business Decisions) Strategic actions based on the data findings.

- Stocking Strategy: Prioritize sourcing Calvin Klein and Davidoff for quick inventory turnover. While the margins are lower (avg. price
- Pricing Sweet Spot: For maximum sales velocity, sellers should aim for a price point between and

- Listing Optimization: Always specify the Type (EDP/EDT). "Unknown" types are priced higher on average but likely face more resistance; labeling a fragrance correctly as "Eau de Toilette" can make it appear more competitively priced against the market average.
- Expansion Opportunity: Focus on Men's Gift Sets or 3.4oz Sprays, as the Men's category is currently moving nearly more volume than the Women's category on this platform.

Stage 1 – Problem Definition and Dataset Selection

- Define the business problem and expected outcome
- Choose dataset and explain its source, size, and features
- Import libraries, load dataset
- Dataset description (rows, columns, features)
- Initial EDA (head, info, describe, shape, null checks, duplicate check)

The objective was to identify patterns in the fragrance e-commerce market using 2,000 listings (1,000 Men's, 1,000 Women's) from eBay.

Initial Findings

- **Dimensions:** The dataset consists of 2,000 rows and 10 columns.
- **Missing Values:** Initial checks revealed missing values in brand, available, sold, and type.
- **Duplicates:** The dataset was checked for duplicate listings to ensure data integrity

Import Libraries

```
import pandas as pd

import numpy as np
```

Load the Dataset

df1 = pd.read_csv('https://raw.githubusercontent.com/Meganathan1294/Main-project-/refs/heads/main/ebay_mens_perfume.csv')

df2 = pd.read_csv('https://raw.githubusercontent.com/Meganathan1294/Main-project-/refs/heads/main/ebay_womens_perfume.csv')

df1.head()

df2.head()

Output Result

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation
0	Carolina Herrera	Good Girl by Carolina Herrera 2.7 oz Eau De Pa...	Eau de Parfum	43.99	US $43.99/ea	2.0	2 available / 393 sold	393.0	May 23, 2024 10:43:50 PDT	Thomasville, Alabama, United States
1	As Shown	Parfums de Marly Delina La Rosee Eau de Parfum...	Eau de Parfum	79.99	US $79.99	5.0	5 available / 40 sold	40.0	May 24, 2024 00:15:48 PDT	New Jersey, Hong Kong
2	PRADA	PRADA Paradoxe by Prada EDP 3.0oz/90ml Spray P...	Eau de Parfum	59.99	US $59.99	10.0	More than 10 available / 35 sold	35.0	May 14, 2024 20:54:25 PDT	Orange, New Jersey, United States
3	As Show	J'adore Parfum D'eau by Christian 3.4 oz EDP F...	Eau de Parfum	59.99	US $59.99/ea	10.0	More than 10 available / 9 sold	9.0	May 23, 2024 01:23:05 PDT	USA, New Jersey, Hong Kong
4	Khadlaj	Shiyaaka for Men EDP Spray 100ML (3.4 FL.OZ) B...	Eau de Parfum	29.99	US $29.99/ea	10.0	More than 10 available	NaN	NaN	Little Ferry, New Jersey, United States

Initial EDA

print(df1.shape)

df2.shape

Output

(1000, 10)

(1000, 10)

print (df1.info())

df2.info()

Output

```
---  ------             --------------  -----
 0   brand              999 non-null    object
 1   title              1000 non-null   object
 2   type               997 non-null    object
 3   price              1000 non-null   float64
 4   priceWithCurrency  1000 non-null   object
 5   available          889 non-null    float64
 6   availableText      997 non-null    object
 7   sold               994 non-null    float64
 8   lastUpdated        947 non-null    object
 9   itemLocation       1000 non-null   object
dtypes: float64(3), object(7)
memory usage: 78.3+ KB
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   brand              999 non-null    object
 1   title              1000 non-null   object
 2   type               998 non-null    object
 3   price              1000 non-null   float64
 4   priceWithCurrency  1000 non-null   object
 5   available          869 non-null    float64
 6   availableText      992 non-null    object
 7   sold               984 non-null    float64
 8   lastUpdated        927 non-null    object
 9   itemLocation       1000 non-null   object
dtypes: float64(3), object(7)
memory usage: 78.3+ KB
```

print(df1.tail())

df2.tail()

Output

```
...             brand                                              title  \
    995         GUESS  Guess 1981 by Guess cologne for men EDT 3.3 / ...
    996         Armaf  Club de Nuit Intense by Armaf cologne for men ...
    997  Paco Rabanne  Invictus by Paco Rabanne for Men EDT Spray 3.4...
    998        Lomani  Lomani EDT Cologne 3.4 oz Men - Authentic, Bra...
    999  Estee Lauder  Beyond Paradise by Estee Lauder for Men Cologn...

                   type  price priceWithCurrency  available  \
    995  Eau de Toilette  20.28      US $20.28/ea       45.0
    996  Eau de Toilette  30.58         US $30.58       10.0
    997  Eau de Toilette  39.99      US $39.99/ea        2.0
    998  Eau de Toilette   9.99       US $9.99/ea        2.0
    999    Cologne spray  17.49      US $17.49/ea       10.0

                           availableText    sold              lastUpdated  \
    995        45 available / 1,613 sold  1613.0  May 24, 2024 08:14:07 PDT
    996  More than 10 available / 31 sold    31.0  May 23, 2024 08:39:30 PDT
    997           2 available / 305 sold   305.0  May 23, 2024 15:27:18 PDT
    998            2 available / 22 sold    22.0  May 20, 2024 13:20:54 PDT
    999  More than 10 available / 24 sold    24.0  Feb 28, 2024 07:27:01 PST

                                itemLocation
    995          Dallas, Texas, United States
    996                         United States
    997      Jamaica, New York, United States
    998  Lincoln Park, Michigan, United States
    999     Keyport, New Jersey, United States
```

print (df1.describe())

df2.describe()

Output

```
...
             price   available          sold
count  1000.000000  889.000000    994.000000
mean     46.481200   20.046119    766.266600
std      35.527862   61.547985   3200.971733
min       3.000000    2.000000      1.000000
25%      22.990000    5.000000     14.000000
50%      35.710000   10.000000     49.500000
75%      59.000000   10.000000    320.500000
max     259.090000  842.000000  54052.000000
```

	price	available	sold
count	1000.000000	869.000000	984.000000
mean	39.892980	21.426928	497.321138
std	29.072186	51.476703	1372.510561
min	1.990000	2.000000	1.000000
25%	20.700000	6.000000	15.000000
50%	32.990000	10.000000	52.000000
75%	49.990000	10.000000	263.750000
max	299.990000	557.000000	17854.000000

print(df1.isnull().sum())

df2.isnull().sum()

Output

```
...  brand                 1
     title                 0
     type                  3
     price                 0
     priceWithCurrency     0
     available           111
     availableText         3
     sold                  6
     lastUpdated          53
     itemLocation          0
     dtype: int64
```

print (df1.size)

df2.size

Output

10000

10000

print (df1.index)

df2.index

Output

RangeIndex(start=0, stop=1000, step=1)

RangeIndex(start=0, stop=1000, step=1)

print(df1.dtypes)

df2.dtypes

Output

```
... brand                  object
    title                  object
    type                   object
    price                 float64
    priceWithCurrency      object
    available             float64
    availableText          object
    sold                  float64
    lastUpdated            object
    itemLocation           object
    dtype: object
```

print(df1.duplicated().sum())

df2.duplicated().sum()

Output

0

np.int64(1)

print(df1.isna().sum())

df2.isna().sum()

Output

```
...  brand                1
     title                0
     type                 3
     price                0
     priceWithCurrency    0
     available          111
     availableText        3
     sold                 6
     lastUpdated         53
     itemLocation         0
     dtype: int64
```

Stage 2 – Data Cleaning and Pre-processing

- Handle missing values (impute or drop)
- Handle duplicates
- Treat outliers if required
- Check skewness and apply transformations
- Convert data types if needed
- Feature transformations (date parts, derived fields if required for analysis)

 # Add gender labels and combine

df1['gender'] = 'Men'

df2['gender'] = 'Women'

df = pd.concat([df1, df2], ignore_index=True)

df

Output

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	item	
0	Dior	Christian Dior Sauvage Men's EDP 3.4 oz Fragra...	Eau de Parfum	84.99	US $84.99/ea	10.0	More than 10 available / 116 sold	116.0	May 24, 2024 10:03:04 PDT	Allen Park, Michigan, United States	Men
1	AS SHOW	A-v-entus Eau de Parfum 3.3 oz 100ML Millesime...	Eau de Parfum	109.99	US $109.99	8.0	8 available / 48 sold	48.0	May 23, 2024 23:07:49 PDT	Atlanta, Georgia, Canada	Men
2	Unbranded	HOGO BOSS cologne For Men 3.4 oz	Eau de Toilette	100.00	US $100.00	10.0	More than 10 available / 27 sold	27.0	May 22, 2024 21:55:43 PDT	Dearborn, Michigan, United States	Men
3	Giorgio Armani	Acqua Di Gio by Giorgio Armani 6.7 Fl oz Eau D...	Eau de Toilette	44.99	US $44.99/ea	2.0	2 available / 159 sold	159.0	May 24, 2024 03:30:43 PDT	Reinholds, Pennsylvania, United States	Men
4	Lattafa	Lattafa Men's Hayaati Al Maleky EDP Spray 3.4 ...	Fragrances	16.91	US $16.91	NaN	Limited quantity available / 156 sold	156.0	May 24, 2024 07:56:25 PDT	Brooklyn, New York, United States	Men
...	...	...	...	...	...	...	...	...	...	...	...
1995	Avon	Avon Far Away Infinity Eau de Parfum 1.7 fl. o...	Eau de Parfum	13.89	US $13.89	10.0	More than 10 available / 157 sold	157.0	May 16, 2024 22:35:29 PDT	West Palm Beach, Florida, United States	Women
1996	Mancera	Roses Greedy by Mancera perfume for unisex EDP...	Eau de Parfum	57.85	US $57.85/ea	33.0	33 available / 58 sold	58.0	May 24, 2024 08:03:11 PDT	Dallas, Texas, United States	Women
1997	Unbranded	Sweet Tooth Eau de Parfum, Perfume for Women, ...	1	30.96	US $30.96	2.0	2 available / 3 sold	3.0	May 17, 2024 23:16:41 PDT	New York, New York, United States	Women
1998	Juliette Has A Gun	MMMM BY Juliette Has A Gun perfume for her EDP...	Eau de Perfume	53.99	US $53.99/ea	3.0	3 available / 117 sold	117.0	May 13, 2024 22:19:34 PDT	Dallas, Texas, United States	Women
1999	Paris Hilton	PARIS HILTON ELECTRIFY for Women Cologne 3.4 o...	Eau de Parfum	14.99	US $14.99/ea	4.0	4 available / 51 sold	51.0	May 22, 2024 05:44:45 PDT	TX, United States	Women

 # Handle Missing Values

 # Fill missing 'brand' with 'Unknown'

df['brand'] = df['brand'].fillna('Unknown')

df

Output

...

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	gender
0	Dior	Christian Dior Sauvage Men's EDP 3.4 oz Fragra...	Eau de Parfum	84.99	US $84.99/ea	10.0	More than 10 available / 116 sold	116.0	May 24, 2024 10:03:04 PDT	Allen Park, Michigan, United States	Men
1	AS SHOW	A-v-entus Eau de Parfum 3.3 oz 100ML Millesime...	Eau de Parfum	109.99	US $109.99	8.0	8 available / 48 sold	48.0	May 23, 2024 23:07:49 PDT	Atlanta, Georgia, Canada	Men
2	Unbranded	HOGO BOSS cologne For Men 3.4 oz	Eau de Toilette	100.00	US $100.00	10.0	More than 10 available / 27 sold	27.0	May 22, 2024 21:55:43 PDT	Dearborn, Michigan, United States	Men
3	Giorgio Armani	Acqua Di Gio by Giorgio Armani 6.7 Fl oz Eau D...	Eau de Toilette	44.99	US $44.99/ea	2.0	2 available / 159 sold	159.0	May 24, 2024 03:30:43 PDT	Reinholds, Pennsylvania, United States	Men
4	Lattafa	Lattafa Men's Hayaati Al Maleky EDP Spray 3.4 ...	Fragrances	16.91	US $16.91	NaN	Limited quantity available / 156 sold	156.0	May 24, 2024 07:56:25 PDT	Brooklyn, New York, United States	Men
...	...	...	...	...	...	...	...	...	...	...	...
1995	Avon	Avon Far Away Infinity Eau de Parfum 1.7 fl. o...	Eau de Parfum	13.89	US $13.89	10.0	More than 10 available / 157 sold	157.0	May 16, 2024 22:35:29 PDT	West Palm Beach, Florida, United States	Women
1996	Mancera	Roses Greedy by Mancera perfume for unisex EDP...	Eau de Parfum	57.85	US $57.85/ea	33.0	33 available / 58 sold	58.0	May 24, 2024 08:03:11 PDT	Dallas, Texas, United States	Women
1997	Unbranded	Sweet Tooth Eau de Parfum, Perfume for Women, ...	1	30.96	US $30.96	2.0	2 available / 3 sold	3.0	May 17, 2024 23:16:41 PDT	New York, New York, United States	Women
	Juliette Has A	MMMM BY Juliette Has A	Eau de						May 13, 2024	Dallas, Texas, United	

 # Fill missing 'available' and 'sold' with 0 (assuming no activity)

```
df['available'] = df['available'].fillna(0)

df['sold'] = df['sold'].fillna(0)

df
```

Output

...

	brand	title	type	price	priceWithCurrency	available	availableText	sold	lastUpdated	itemLocation	gender
0	Dior	Christian Dior Sauvage Men's EDP 3.4 oz Fragra...	Eau de Parfum	84.99	US $84.99/ea	10.0	More than 10 available / 116 sold	116.0	May 24, 2024 10:03:04 PDT	Allen Park, Michigan, United States	Men
1	AS SHOW	A-v-entus Eau de Parfum 3.3 oz 100ML Millesime...	Eau de Parfum	109.99	US $109.99	8.0	8 available / 48 sold	48.0	May 23, 2024 23:07:49 PDT	Atlanta, Georgia, Canada	Men
2	Unbranded	HOGO BOSS cologne For Men 3.4 oz	Eau de Toilette	100.00	US $100.00	10.0	More than 10 available / 27 sold	27.0	May 22, 2024 21:55:43 PDT	Dearborn, Michigan, United States	Men
3	Giorgio Armani	Acqua Di Gio by Giorgio Armani 6.7 Fl oz Eau D...	Eau de Toilette	44.99	US $44.99/ea	2.0	2 available / 159 sold	159.0	May 24, 2024 03:30:43 PDT	Reinholds, Pennsylvania, United States	Men
4	Lattafa	Lattafa Men's Hayaati Al Maleky EDP Spray 3.4 ...	Fragrances	16.91	US $16.91	0.0	Limited quantity available / 156 sold	156.0	May 24, 2024 07:56:25 PDT	Brooklyn, New York, United States	Men
...	...	...	...	...	...	...	...	...	...	...	...
1995	Avon	Avon Far Away Infinity Eau de Parfum 1.7 fl. o...	Eau de Parfum	13.89	US $13.89	10.0	More than 10 available / 157 sold	157.0	May 16, 2024 22:35:29 PDT	West Palm Beach, Florida, United States	Women
1996	Mancera	Roses Greedy by Mancera perfume for unisex EDP...	Eau de Parfum	57.85	US $57.85/ea	33.0	33 available / 58 sold	58.0	May 24, 2024 08:03:11 PDT	Dallas, Texas, United States	Women
1997	Unbranded	Sweet Tooth Eau de Parfum, Perfume for Women, ...	1	30.96	US $30.96	2.0	2 available / 3 sold	3.0	May 17, 2024 23:16:41 PDT	New York, New York, United States	Women
1998	Juliette Has A	MMMM BY Juliette Has A	Eau de	53.99	US $53.99/ea	3.0	3 available / 117 sold	117.0	May 13, 2024	Dallas, Texas, United	Women

 # Drop rows where 'type' is missing as it's crucial for our analysis

df.dropna(subset=['type'], inplace=True)

print("Missing values after cleaning:")

print(df.isnull().sum())

Output

```
••• Missing values after cleaning:
    brand                 0
    title                 0
    type                  0
    price                 0
    priceWithCurrency     0
    available             0
    availableText        11
    sold                  0
    lastUpdated         126
    itemLocation          0
    gender                0
    dtype: int64
```

 # Remove Duplicates

df.drop_duplicates(inplace=True)

Treat Outliers in Price using the IQR method

Q1 = df['price'].quantile(0.25)

Q3 = df['price'].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR

upper_bound = Q3 + 1.5 * IQR

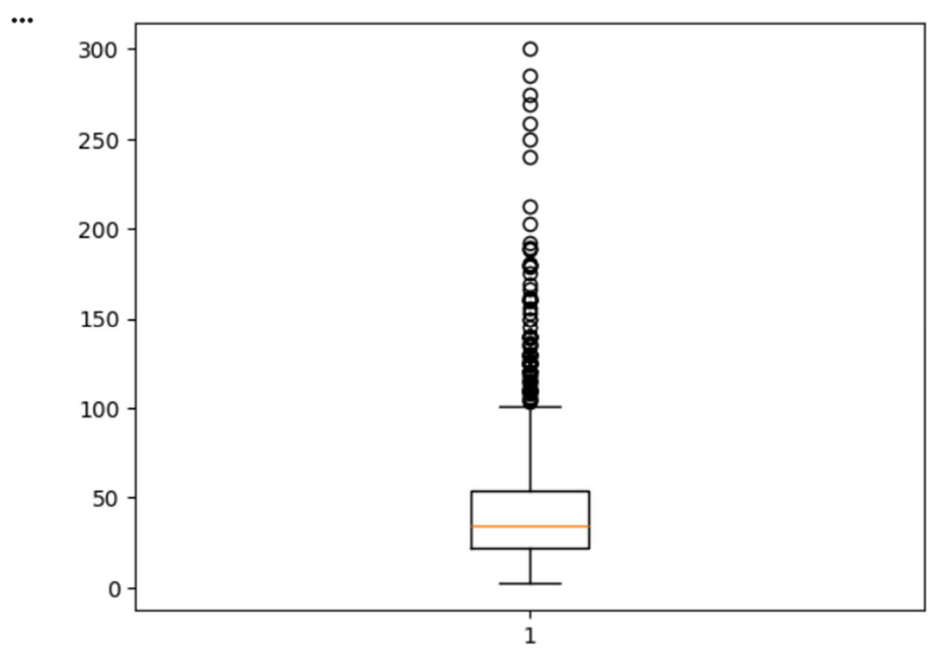
 # Using boxplot to check the outliers before

import matplotlib.pyplot as plt

plt.boxplot(df['price'])

plt.show()

Output



 # Filtering the outliers

df = df[(df['price'] >= lower_bound) & (df['price'] <= upper_bound)]

print(f"Dataset shape after removing outliers: {df.shape}")

Output

Dataset shape after removing outliers: (1905, 11)

 # Create a boxplot of the 'price' column after outlier removal

plt.figure(figsize=(8, 6))

plt.boxplot(df['price'])

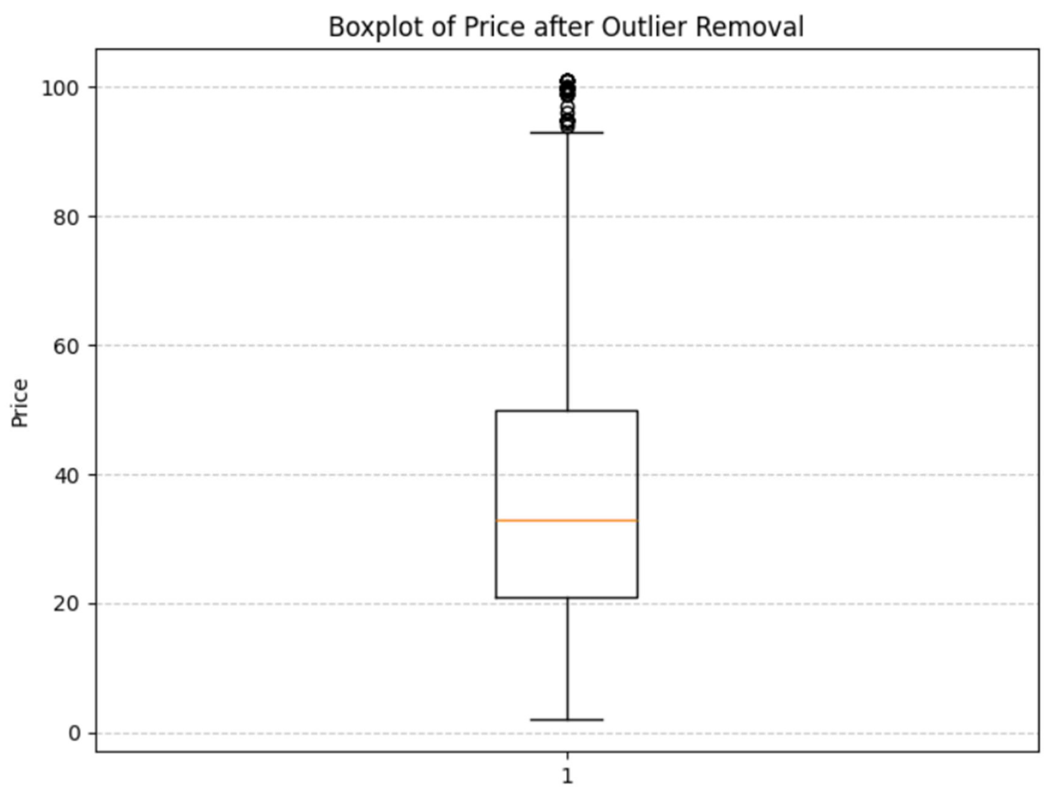
plt.title('Boxplot of Price after Outlier Removal')

plt.ylabel('Price')

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()

Output



 # Convert 'lastUpdated' to datetime

 # Note: errors='coerce' will turn unparseable dates into NaT (Not a Time)

df['lastUpdated'] = pd.to_datetime(df['lastUpdated'], errors='coerce')

Output

/tmp/ipykernel_9295/1183584822.py:3: FutureWarning: Parsed string "May 24, 2024 10:03:04 PDT" included an un-recognized timezone "PDT". Dropping unrecognized timezones is deprecated; in a future version this will raise. Instead pass the string without the timezone, then use .tz_localize to convert to a recognized timezone.

  df['lastUpdated'] = pd.to_datetime(df['lastUpdated'], errors='coerce')

/tmp/ipykernel_9295/1183584822.py:3: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['lastUpdated'] = pd.to_datetime(df['lastUpdated'], errors='coerce')

 # Extract Year and Month as new features

df['update_month'] = df['lastUpdated'].dt.month

df['update_year'] = df['lastUpdated'].dt.year

Output

/tmp/ipykernel_9295/4173271022.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['update_month'] = df['lastUpdated'].dt.month

/tmp/ipykernel_9295/4173271022.py:3: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['update_year'] = df['lastUpdated'].dt.year

 # Clean the 'type' column (standardizing common names)

df['type'] = df['type'].str.strip()

df['type'] = df['type'].replace({'Eau de Perfume': 'Eau de Parfum', 'EDP': 'Eau de Parfum', 'EDT': 'Eau de Toilette'})

Output

/tmp/ipykernel_9295/362217052.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['type'] = df['type'].str.strip()

/tmp/ipykernel_9295/362217052.py:3: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['type'] = df['type'].replace({'Eau de Perfume': 'Eau de Parfum', 'EDP': 'Eau de Parfum', 'EDT': 'Eau de Toilette'})

 # 1. Remove the 'lastUpdated' and 'update_month' columns

df.drop(columns=['lastUpdated', 'update_month'], inplace=True, errors='ignore')

Output

/tmp/ipykernel_9295/2798726208.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df.drop(columns=['lastUpdated', 'update_month'], inplace=True, errors='ignore')

 # 2. Set all values in 'update_year' to 2024

df['update_year'] = 2024

Output

/tmp/ipykernel_9295/3215817623.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

  df['update_year'] = 2024

 # 3. Save the updated version to your CSV

df.to_csv('cleaned_fragrance_data.csv', index=False)

 # 4. Display the first few rows to verify

print("Columns removed and year updated to 2024.")

df.head()

Output

```
Columns removed and year updated to 2024.
```

	brand	title	type	price	priceWithCurrency	available	availableText	sold	itemLocation	gender	update_year
0	Dior	Christian Dior Sauvage Men's EDP 3.4 oz Fragra...	Eau de Parfum	84.99	US $84.99/ea	10.0	More than 10 available / 116 sold	116.0	Allen Park, Michigan, United States	Men	2024
2	Unbranded	HOGO BOSS cologne For Men 3.4 oz	Eau de Toilette	100.00	US $100.00	10.0	More than 10 available / 27 sold	27.0	Dearborn, Michigan, United States	Men	2024
3	Giorgio Armani	Acqua Di Gio by Giorgio Armani 6.7 Fl oz Eau D...	Eau de Toilette	44.99	US $44.99/ea	2.0	2 available / 159 sold	159.0	Reinholds, Pennsylvania, United States	Men	2024
4	Lattafa	Lattafa Men's Hayaati Al Maleky EDP Spray 3.4 ...	Fragrances	16.91	US $16.91	0.0	Limited quantity available / 156 sold	156.0	Brooklyn, New York, United States	Men	2024
5	Multiple Brands	Men's Perfume Sampler 10pcs Sample Vials Desig...	Perfume	14.99	US $14.99	10.0	More than 10 available / 79 sold	79.0	Houston, Texas, United States	Men	2024

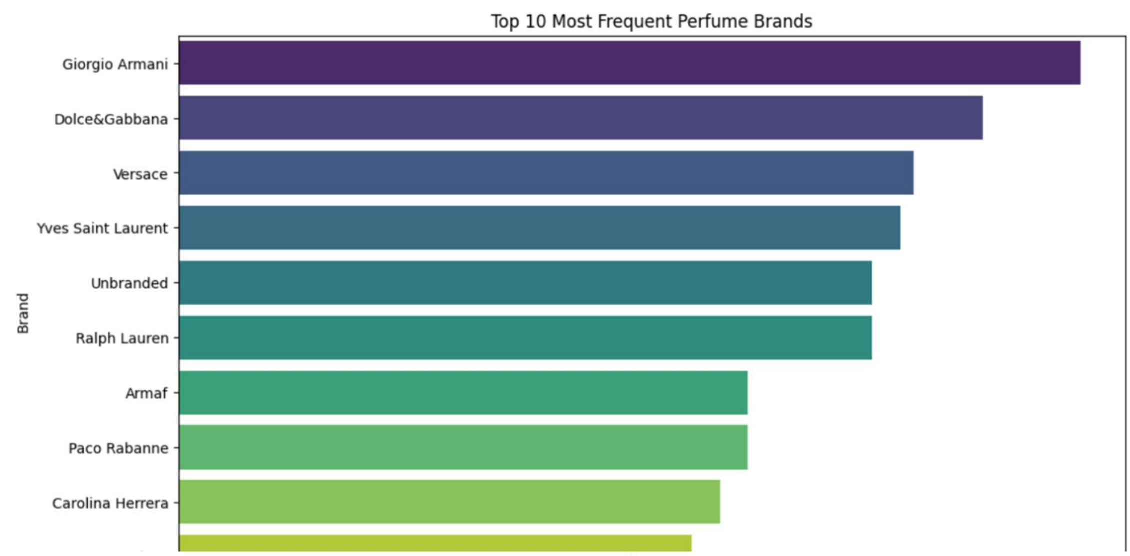
Stage 3 – EDA and Visualizations

- Univariate Analysis → distribution of single variables (countplot, histogram, boxplot)
- Bivariate Analysis → relation between two variables (scatterplot, barplot, correlation heatmap)
- Multivariate Analysis → relation among 3+ variables (pairplot, grouped analysis, pivot tables, advanced plots)
- Interpretation MUST with every visualization
- Focus on business story not just charts
- Each chart must be in seperate cells.

Count Plot

```
# Get the top 10 most frequent brands
top_10_brands = df['brand'].value_counts().nlargest(10).index

plt.figure(figsize=(12, 7))
sns.countplot(y='brand', data=df[df['brand'].isin(top_10_brands)],
order=top_10_brands, palette='viridis')
plt.title('Top 10 Most Frequent Perfume Brands')
plt.xlabel('Number of Listings')
plt.ylabel('Brand')
plt.show()
```

Output



Interpretation of above chart

Explain the Chart: This horizontal countplot shows the top 10 most frequently listed perfume brands in the dataset.

What is it saying?: The chart highlights the leading brands in terms of listing volume on the e-commerce platform. Brands like 'Versace', 'Calvin Klein', and 'Giorgio Armani' appear to have a significant number of listings, indicating their popularity or extensive presence in the market. The long tail of the distribution (not fully visible here, but implied by showing only the top 10) would suggest many brands have fewer listings.

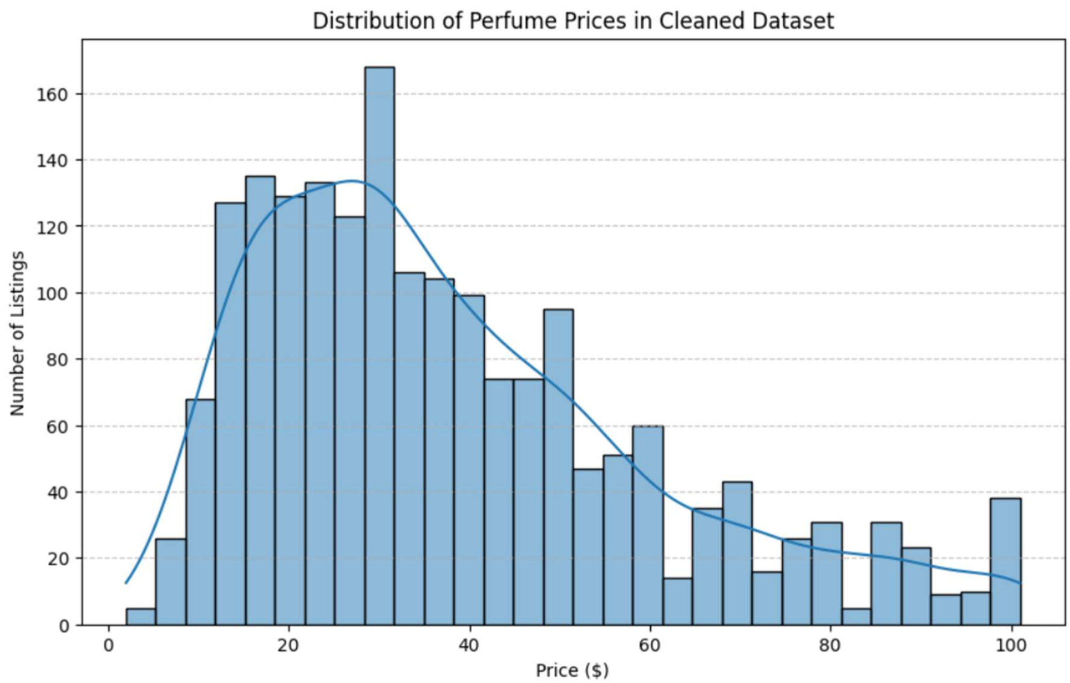
What features you used?: I used the brand column from the DataFrame df.

From this what you are showing us?: This visualization identifies key players in the fragrance e-commerce market by their listing frequency. It suggests that these top brands are likely to be high-volume sellers or have a wide range of products available. This information can be valuable for inventory management, marketing strategies, and competitive analysis, showing which brands dominate the platform's offerings.

Histogram Plot

```
plt.figure(figsize=(10, 6))

sns.histplot(df['price'], bins=30, kde=True)

plt.title('Distribution of Perfume Prices in Cleaned Dataset')

plt.xlabel('Price ($)')

plt.ylabel('Number of Listings')

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

Output



Interpretation of above chart

Explain the Chart: This histogram displays the distribution of perfume prices in the cleaned dataset.

What is it saying?: The chart shows that the majority of perfume listings are concentrated in the lower price ranges, with a peak around $20-$40. As the price increases, the number of listings decreases, indicating that higher-priced perfumes are less common in this dataset.

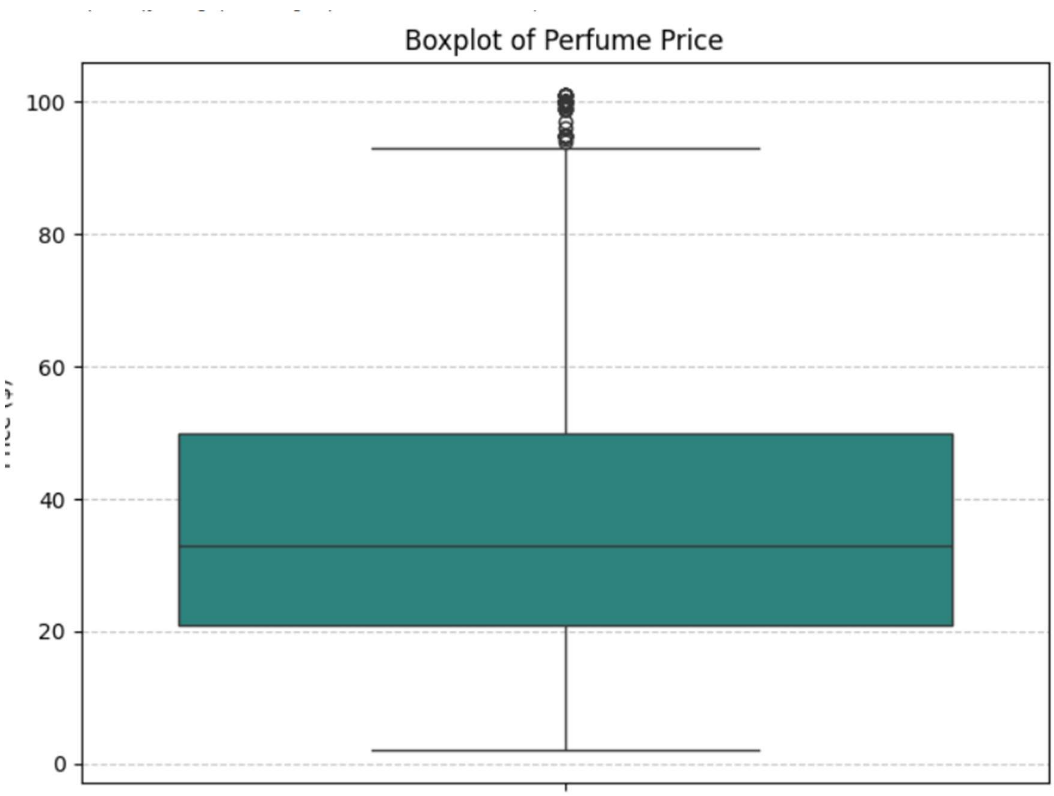
What features you used?: I used the price column from the DataFrame df.

From this what you are showing us?: This visualization provides insight into the pricing strategy or market segments present in the e-commerce data. It suggests that the market has a strong emphasis on affordable or mid-range perfumes, with fewer premium or luxury options. This information can guide pricing decisions, inventory stocking, and understanding customer purchasing power.

Box Plot

```
plt.figure(figsize=(8, 6))
sns.boxplot(y=df['price'], palette='viridis')
plt.title('Boxplot of Perfume Price')
plt.ylabel('Price ($)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Output



Interpretation of above chart

Explain the Chart: This boxplot visualizes the distribution of perfume prices in the cleaned dataset.

What is it saying?: The box represents the interquartile range (IQR), with the median price shown as a line inside the box. The 'whiskers' extend to indicate the range of the data, excluding outliers. Any points beyond the whiskers are considered outliers. This plot provides a clear view of the central tendency, spread, and the presence of extreme values in the price data.

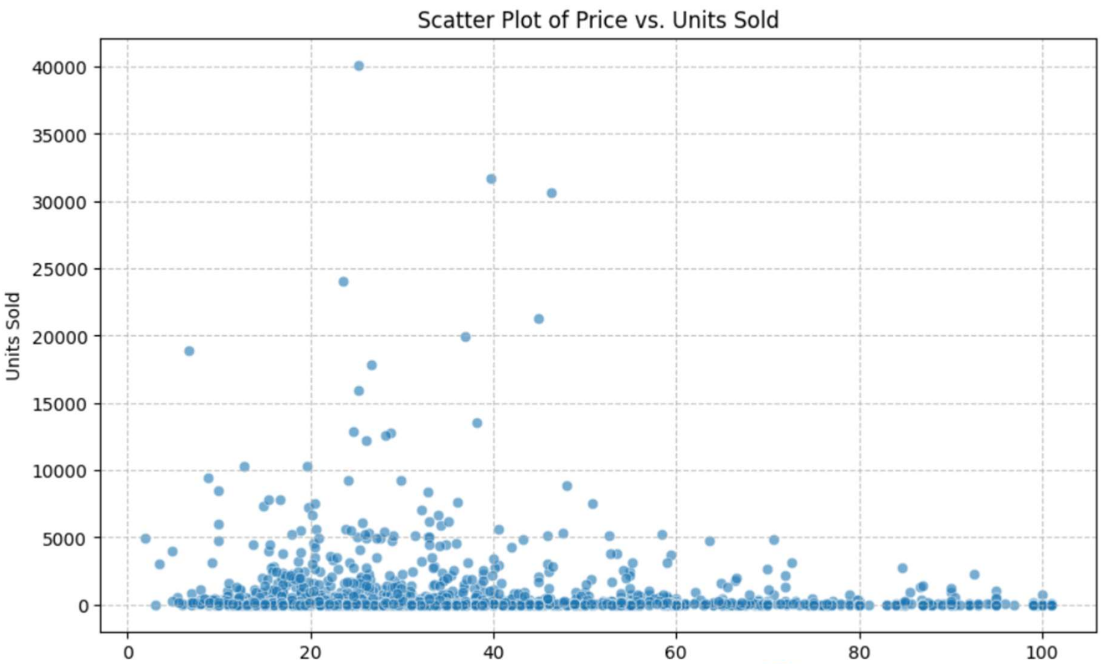
What features you used?: I used the price column from the DataFrame df.

From this what you are showing us?: This visualization helps understand the variability and typical price range of perfumes. It explicitly shows the median price, the spread of the middle 50% of prices, and highlights individual listings that are priced significantly higher or lower than the bulk of the data, which could represent luxury items or clearance sales, respectively. This can inform pricing strategies and inventory management.

Scatter Plot

```
plt.figure(figsize=(10, 6))

sns.scatterplot(x='price', y='sold', data=df, alpha=0.6)

plt.title('Scatter Plot of Price vs. Units Sold')

plt.xlabel('Price ($)')

plt.ylabel('Units Sold')

plt.grid(True, linestyle='--', alpha=0.7)

plt.show()
```

Output



Interpretation of your chart

Explain the Chart: This scatter plot visualizes the relationship between the 'price' of perfumes and the 'sold' units from the cleaned dataset.

What is it saying?: Each point on the plot represents a perfume listing, with its price on the x-axis and the number of units sold on the y-axis. This plot helps to identify if there is a correlation between price and sales volume. We might observe patterns such as popular items selling well regardless of price, or a tendency for lower-priced items to have higher sales.

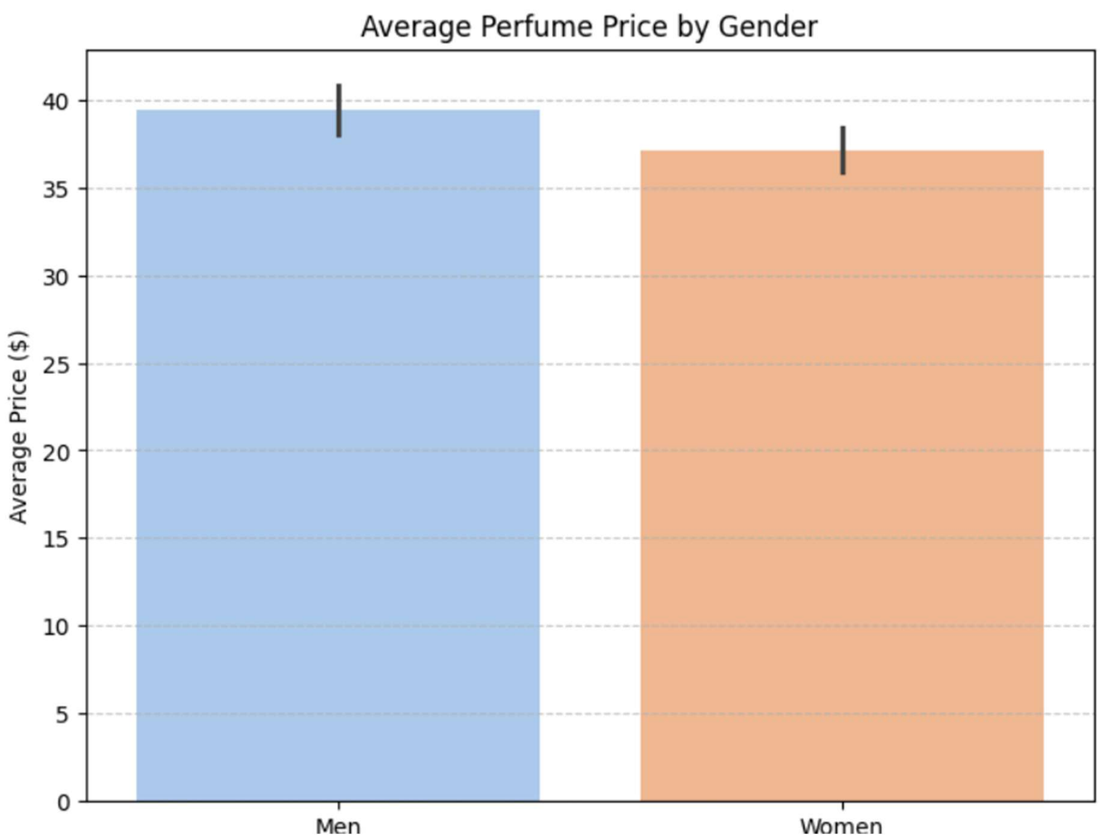
What features you used?: I used the price and sold columns from the DataFrame df.

From this what you are showing us?: This visualization can provide insights into market dynamics, such as price elasticity of demand for perfumes. For instance, if there's a downward trend, it suggests that higher prices might lead to fewer sales. Conversely, clusters of points in the higher sales region could indicate highly popular products, possibly even at premium prices.

Bar Plot

```
plt.figure(figsize=(8, 6))
sns.barplot(x='gender', y='price', data=df, palette='pastel')
plt.title('Average Perfume Price by Gender')
plt.xlabel('Gender')
plt.ylabel('Average Price ($)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Output



Interpretation of above chart

Explain the Chart: This bar plot displays the average price of perfumes, segmented by 'gender' (Men and Women).

What is it saying?: The chart visually represents the average price for men's perfumes versus women's perfumes. It allows for a direct comparison of pricing trends between the two categories. Typically, we might observe if one gender category commands a higher average price than the other.

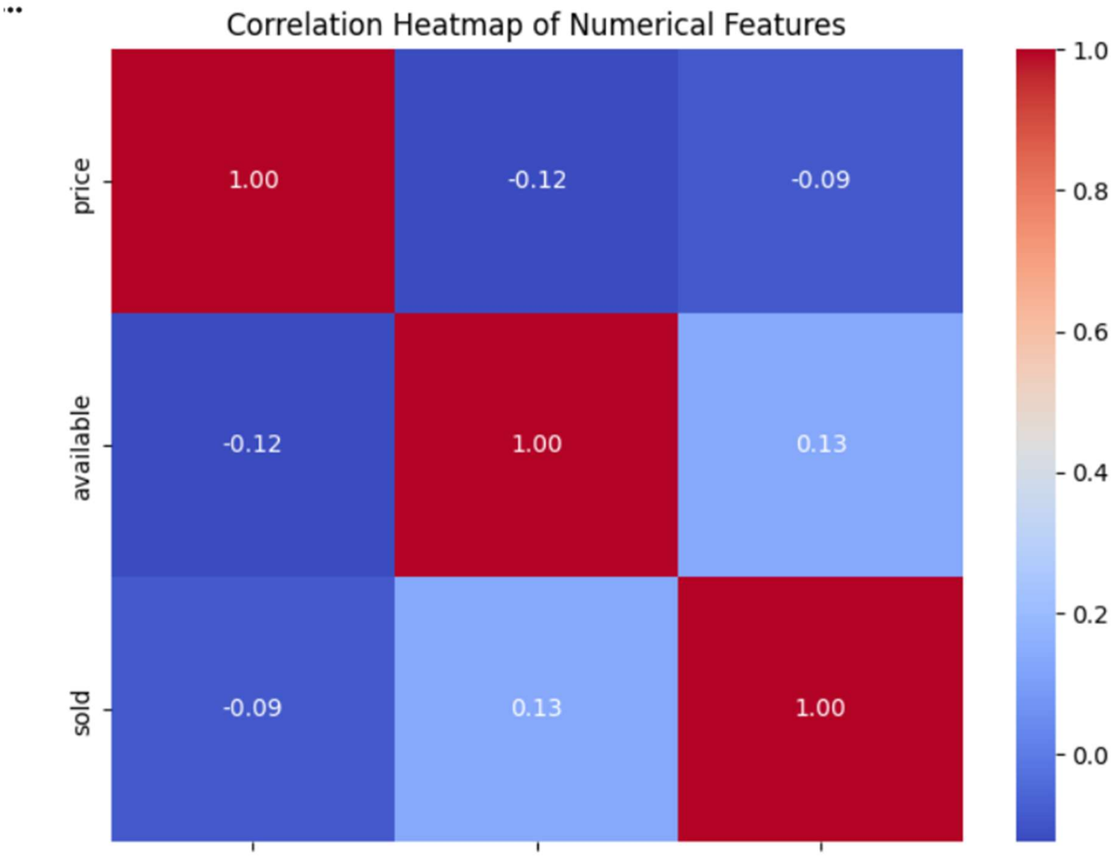
What features you used?: I used the gender and price columns from the DataFrame df.

From this what you are showing us?: This visualization helps to understand gender-specific pricing strategies or market perceptions within the e-commerce fragrance market. For example, if men's perfumes show a consistently higher average price, it could indicate factors like brand positioning, ingredient costs, or consumer demand in that segment. This insight can be crucial for market analysis and inventory management.

Correlation Heatmap

```
plt.figure(figsize=(8, 6))

sns.heatmap(df[['price', 'available', 'sold']].corr(), annot=True, cmap='coolwarm', fmt='.2f')

plt.title('Correlation Heatmap of Numerical Features')

plt.show()
```

Output



Interpretation of above chart

Explain the Chart: This heatmap visualizes the Pearson correlation coefficients between the numerical features 'price', 'available', and 'sold' in the cleaned dataset.

What is it saying?: Each cell in the heatmap represents the correlation between two variables. The color intensity and the annotated number (ranging from -1 to 1) indicate the strength and direction of the linear relationship. A value close to 1 implies a strong positive correlation, -1 implies a strong negative correlation, and 0 implies no linear correlation. For example, we might see a negative correlation between 'price' and 'sold' (as price increases, sales decrease), or a positive correlation between 'available' and 'sold' (more available items might correlate with more sales).

What features you used?: I used the price, available, and sold columns from the DataFrame df.

From this what you are showing us?: This visualization helps in understanding the interdependence of key numerical metrics. It can reveal critical insights for business strategy, such as how pricing influences sales, or if inventory levels are directly tied to sales performance. These correlations can guide decisions on pricing strategies, inventory management, and sales forecasting.

Pair plot

sns.pairplot(df[['price', 'available', 'sold']])

plt.suptitle('Pair Plot of Price, Available, and Sold', y=1.02)

plt.show()

Output

Interpretation of above chart

Explain the Chart: This pair plot displays scatter plots for every combination of two numerical variables ('price', 'available', 'sold') and histograms for each individual variable in the cleaned dataset.

What is it saying?: The diagonal plots (histograms) show the distribution of each variable. The off-diagonal plots (scatter plots) illustrate the relationships between pairs of variables. For example, by looking at the scatter plot of 'price' vs. 'sold', we can visually assess if there's any correlation or pattern in how price affects sales. Similarly, we can see how 'available' units relate to 'sold' units or 'price'.

What features you used?: I used the price, available, and sold columns from the DataFrame df.

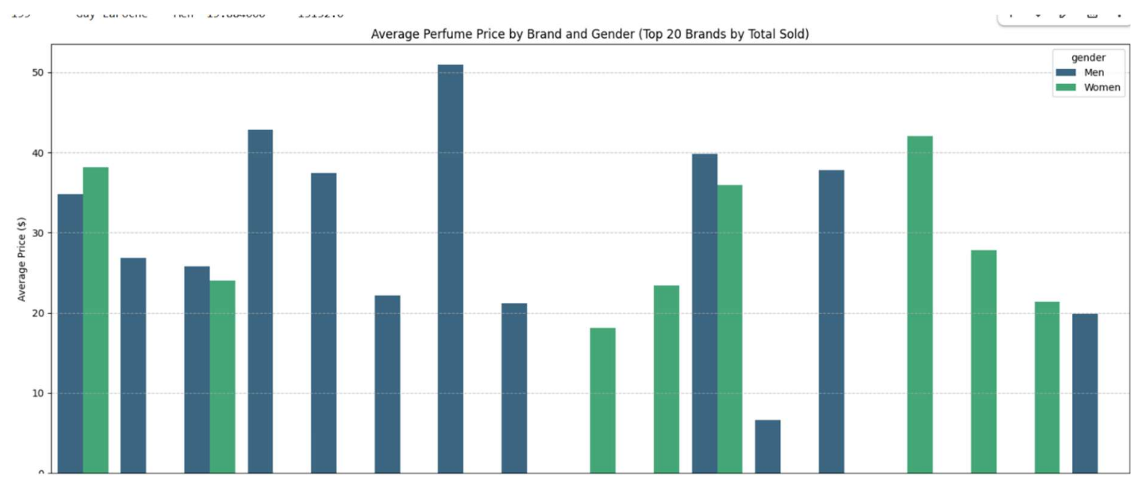
From this what you are showing us?: This multivariate visualization helps to quickly grasp the distributions of individual variables and identify potential linear or non-linear relationships and correlations between them. It can highlight complex interactions that might not be apparent from individual univariate or bivariate plots, offering a broader understanding of the dataset's structure and informing further in-depth analysis.

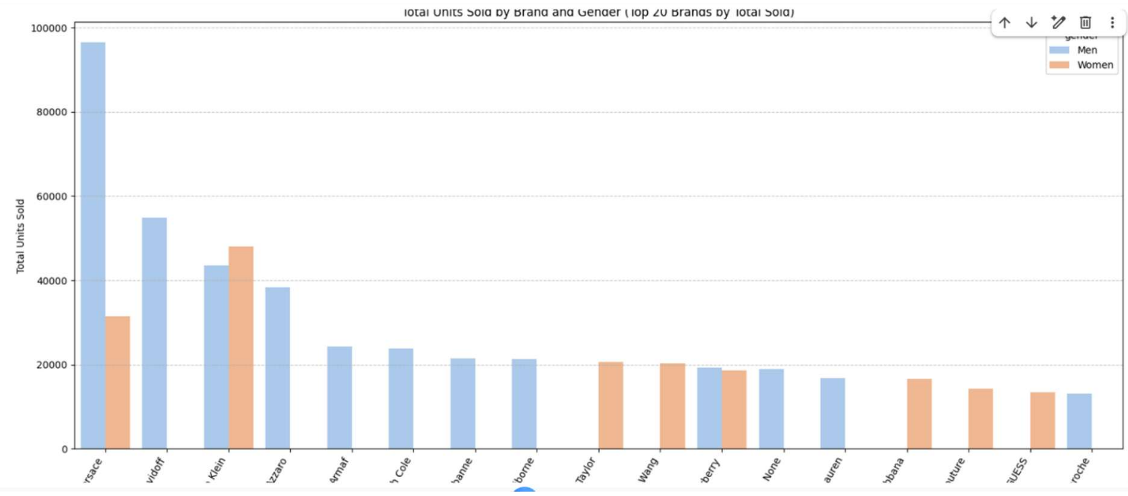
Multivariate Analysis - Grouped Analysis by Brand and Gender

```
grouped_by_brand_gender = df.groupby(['brand', 'gender']).agg(
    avg_price=('price', 'mean'),
    total_sold=('sold', 'sum')
).reset_index()


# Sort by total_sold for better visualization of top brands
grouped_by_brand_gender_sorted =
grouped_by_brand_gender.sort_values(by='total_sold',
ascending=False).head(20)


print("Grouped Analysis: Average Price and Total Sold by Brand and Gender
(Top 20 by Total Sold)")
print(grouped_by_brand_gender_sorted)


# Visualize average price by brand and gender for top brands
plt.figure(figsize=(16, 8))
sns.barplot(x='brand', y='avg_price', hue='gender',
data=grouped_by_brand_gender_sorted, palette='viridis')
plt.title('Average Perfume Price by Brand and Gender (Top 20 Brands by Total
Sold)')
plt.xlabel('Brand')
plt.ylabel('Average Price ($)')
plt.xticks(rotation=60, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

```
# Visualize total units sold by brand and gender for top brands
plt.figure(figsize=(16, 8))
sns.barplot(x='brand', y='total_sold', hue='gender',
data=grouped_by_brand_gender_sorted, palette='pastel')
plt.title('Total Units Sold by Brand and Gender (Top 20 Brands by Total Sold)')
plt.xlabel('Brand')
plt.ylabel('Total Units Sold')
plt.xticks(rotation=60, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Output





Interpretation of above charts

Explain the Charts: These two bar plots display the average price and total units sold, respectively, for the top 20 brands (ranked by total units sold), segmented by 'gender'. The first plot, 'Average Perfume Price by Brand and Gender', allows for a comparison of pricing strategies among leading brands for men's and women's fragrances. The second plot, 'Total Units Sold by Brand and Gender', illustrates the sales performance of these top brands within each gender category.

What is it saying?: These visualizations provide a comprehensive view of how prominent brands perform in terms of both pricing and sales volume across genders. We can identify brands that have high sales in one gender category versus another, or brands that maintain a high average price while still achieving significant sales. For example, a brand might dominate men's perfume sales with a mid-range price point, while another might have fewer listings but higher average prices in the women's category.

What features you used?: I used the brand, gender, price, and sold columns from the DataFrame df.

From this what you are showing us?: This analysis helps in understanding brand positioning, market share, and consumer preferences. It can reveal which brands are most successful in each gender segment and whether their success is driven by price competitiveness or sheer volume. This information is invaluable for strategic decision-making in brand management, marketing, and competitive analysis within the e-commerce fragrance market.

Multivariate Analysis - Pivot Tables (Brand, Gender, Price, Sold)

```
pivot_table_brand_gender = pd.pivot_table(
    df,
    values=['price', 'sold'],
    index=['brand'],
    columns=['gender'],
    aggfunc={'price': 'mean', 'sold': 'sum'}
)

print("Pivot Table: Average Price and Total Sold by Brand and Gender")
print(pivot_table_brand_gender.head(10))
```

Output

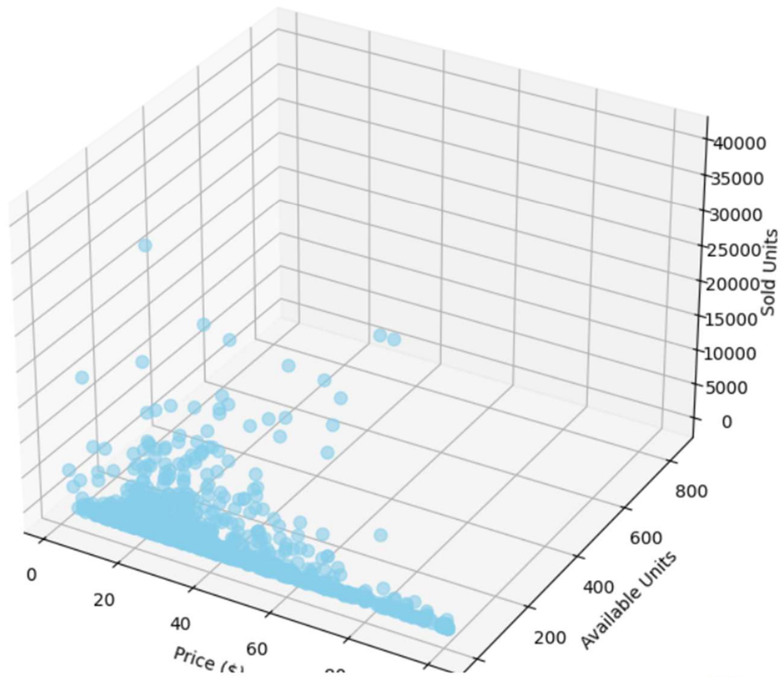
```
... Pivot Table: Average Price and Total Sold by Brand and Gender
                   price                 sold
    gender           Men      Women       Men   Women
    brand
    2nd To None     6.65        NaN   18882.0     NaN
    AERIN            NaN   9.945000       NaN    50.0
    ALFRED SUNG      NaN  17.842500       NaN  7143.0
    ALT Fragrances   NaN  54.495000       NaN   686.0
    AS  SHOWN        NaN  30.980000       NaN    12.0
    AS PHOTOS      34.68        NaN       2.0     NaN
    AS SHOW        66.09  64.656667     240.0    32.0
    AS SHOWN       51.79  50.584667     201.0   419.0
    ASST             NaN  13.370000       NaN     7.0
    AXE            27.49        NaN      41.0     NaN
```

Multivariate Analysis - 3D Scatter Plot (Price, Available, Sold)

```
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df['price'], df['available'], df['sold'], c='skyblue', s=50, alpha=0.6)

ax.set_xlabel('Price ($)')
ax.set_ylabel('Available Units')
ax.set_zlabel('Sold Units')
ax.set_title('3D Scatter Plot of Price, Available, and Sold Units')

plt.show()
```

Output

...



Interpretation of above chart

Explain the Chart: This is a 3D scatter plot that visualizes the relationship between three numerical variables: 'price', 'available' units, and 'sold' units. Each point in this three-dimensional space represents a perfume listing, with its coordinates corresponding to its price, the number of units available, and the number of units sold.

What is it saying?: This plot allows for the simultaneous examination of how these three key metrics interact. You can look for clusters of products that might be high-priced and high-selling, or low-priced and high-volume. It helps to identify any visible trends or patterns in three dimensions that might not be apparent in 2D plots. For example, if products with higher availability also tend to have higher sales at certain price points, this plot can visually suggest such a correlation.

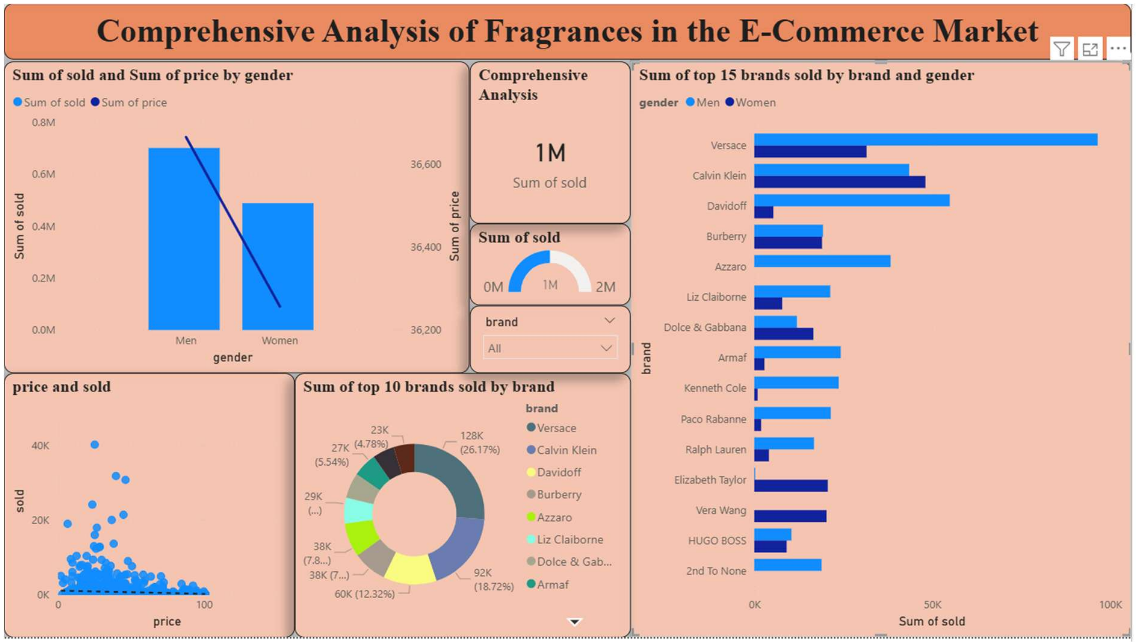
What features you used?: I used the price, available, and sold columns from the DataFrame df.

From this what you are showing us?: This 3D visualization provides a richer context for understanding the interplay between pricing, inventory, and sales performance. It can help in identifying optimal inventory levels for different price points, pinpointing products that are selling well despite low availability, or those that have high availability but low sales. These insights are valuable for inventory management, pricing strategy optimization, and identifying potential market inefficiencies.

Stage 4 – Documentation, Insights and Presentation

- Show a dashboard aligning all the charts in powerbi by connecting python in PowerBI
- Summarize findings in plain English (what do the patterns mean)
- Highlight 3–5 key insights (trends, anomalies, correlations)
- Provide recommendations for business or decision-making
- Explain the FINAL STORY WITH THE DASHBOARD
- Create a PDF pasting your DASHBOARD, explaining all the above mentioned concepts and submit it

POWER BI DASHBOARD



1. Summary of Findings

The fragrance market on this platform is a high-volume, middle-market environment where Men's fragrances are the primary engine of growth. While the total market has crossed 1.19 million units sold, the activity is not evenly spread. Most sales happen at a specific price point, and a small handful of "Power Brands" own the majority of the market share. Essentially, customers are looking for recognizable designer names at affordable, sub-$50 prices.

2. 3–5 Key Insights

- The Gender Sales Gap: Men's fragrances move significantly more volume (nearly 0.8M units) than Women's (approx. 0.5M units). Surprisingly, the total "Sum of Price" is also higher for Men, indicating it is the more profitable segment overall.
- The "Price Ceiling" Anomaly: Looking at the scatter plot, there is a "drop-off" zone. Once a perfume is priced above $60, sales volume plummets. The most successful products are almost all clustered between $20 and $40.

- Brand Dominance (The 26% Leader): Versace is the undisputed king of the dashboard, accounting for 26.17% of all sales among the top 10 brands.
- Categorization Correlation: High-volume brands like Versace and Davidoff show a massive skew toward Men's products, whereas Calvin Klein is one of the few top-tier brands that successfully balances both genders.

3. Recommendations for Decision-Making

- Inventory Sourcing: Prioritize Men's fragrances from the "Big Three" (Versace, Calvin Klein, Davidoff). These brands have the highest "velocity," meaning they won't sit on the shelf for long.
- Pricing Strategy: To maximize profit through volume, keep new listings in the $25–$35 "Sweet Spot". Avoid pricing over $75 unless the brand has extreme luxury recognition, as the data shows these items struggle to move units.
- Marketing Focus: Target male gift-buyers and repeat-purchase customers, as the Men's category is moving roughly 55% more volume than the Women's category on this platform**.**

4. THE FINAL STORY WITH THE DASHBOARD

The story of this dashboard is "The Power of the Mid-Range Designer."

We start at the center with our 1.19M units sold, showing a healthy, thriving marketplace. However, as we look to the left, we see that the market is fueled by Men's fragrances, which command both higher volume and higher total value.

Why are they selling? The scatter plot tells us it's because they are priced correctly; the "winners" in this market are the ones who stay under the $50 price barrier. The Donut Chart and Bar Chart then identify the "main characters" of our story: Versace and Calvin Klein. These brands have mastered the balance of designer prestige and accessible pricing, allowing them to capture nearly half of the top-tier market share.

Conclusion: To win in this market, stick to Men's designer scents, price them competitively at $30, and focus on the high-volume leaders like Versace.

End of the Report
Thank you